



Applying Ballistic Launching Mechanisms to Optimize Transporter-Erector-Launcher (TEL) Performances in Conjunction with Short-Range Ballistic Missiles

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Abstract

The purpose of this investigation is to improve a small-scale base model of a Transporter-Erector-Launcher (TEL) platform for a subsidiary of Dauntless Industries, LLC via an increase in ease of use, efficiency, range of projectile, and accuracy, among other criteria. The scope of this research encompasses TEL design remediation nationwide for efficient Short-Range Ballistic Missile (SRBM) launches.

A Design Brief derived from the Request For Proposal will be used to generate concept sketches of an improved TEL structure, before selecting a design to pursue via a Decision Matrix. The final design, once drafted (both in a 3-view and isometric technical drawing), will be implemented using the VEX V5 Robotics components, and every possible input combination will be repeatedly tested to ensure design accuracy.

The redesigned TEL vehicle for SRBM systems implemented exceeded its target constraints and functioned with precision throughout all trials, successfully demonstrating continuous motor motion throughout launch angles (10-80 degrees), consistent initial velocity through all launch angles, unmanned adjustment of robot motion and aim, and functionality of LED status indicators. This design will increase ballistic launching performance, leading to precision and accuracy in ballistic missile operations, such as autonomous and unmanned trials currently conducted by the U.S. Army. It is recommended that the client use the findings of this investigation to optimize TELs for SRBM systems with an enlarged scheme of the designed solution.



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1.0 Introduction

1.1 Subject

Ballistic Launching Mechanisms are grounded in 2-dimensional Kinematics, and more specifically, Projectile Motion, which is field in Classical Mechanics of Physics. Related terminology will include knowledge of the colloquially-known Big Five Formulas in Kinematics and related constants (e.g. the gravitational constant g).

The first recorded use of Ballistic Launchers was recorded in 1232 in Kaifeng, the capital of the Henan province in China. However, this early use was simply for recreational purposes, such as setting off fireworks and burning old wicker works. With the emergence of European colonial powers, Enlightenment ideals, and technological advancements, the rudimentary rockets of the Chinese evolved into long-range missiles. The usage of Ballistics came to fruition in 1807, when the underpowered British naval fleet launched thousands of iron rockets – early missiles – and destroyed the massive French navy stationed in Copenhagen, Denmark. Furthermore, in Francis Scott Key's *Star Spangled Banner*, he describes seeing the American flag "by the rocket's red glare." To put a cherry on top, German V-2 Ballistic Missiles were produced at the end of World War II, and met with equal efforts from First World Countries, such as the United States' Patriot anti-aircraft system. Evidently, Ballistic Missiles had transformed into the norm of warfare, and have continued to develop into modern-day situations, such as the Russo-Ukrainian War currently ongoing.

The significance of this research investigation lies in its applicability in modern-day society. Missile Ballistics have become the face of modern warfare and encompass a major part in National Security and defense administrations across the world. Hence, this investigation seeks to develop a relatively cost-efficient Ballistic Launching Device that will at least traverse 15 feet so that the design can be adapted to a massive scale and put to use on the frontlines.

1.2 Purpose

The project's purpose is to successfully improve Hobby Spectacle's existing Ballistic Launcher through a series of 5 modifications that will increase ease of use, efficiency, range of projectile, and accuracy, among other criteria. This report entails the several procedures and methods used to successfully design a recycling system able to sort the marbles through both the hardware and software designs. The final design will be given to the Directors of Panther Engineering, who will convey the design specifications to Hobby Spectacle and Dauntless Industries, LLC; an additional review to inspect the hardware and software designs for the finished Ballistic Launcher system by other parties (see Section 2.4: Audience) will take place as well.

1.3 Scope

This report's scope will be limited to the VEX components and materials on hand that do not consider wear-and-tear conditions for the hardware design. With only the smart & servo motors, the only VEX motors in the kit has limited the variety of motors to these two, severely limiting the design for a Ballistic Launching system. Constraints faced on this project include a minimum requirement of 15 feet traveled by the projectile, continuous motion of the robot through angles from 10 degrees to 80 degrees, and a fixed initial velocity for precision purposes.

1.4 Audience



This report is intended for the Directors of Panther Engineering so that the final Ballistic Launching device may be assessed and implemented in Hobby Spectacle's base model, under the jurisdiction of Dauntless Industries, LLC. Alternatively, the findings of this study may be reviewed by the United States Department of Defense and the American Society of Civil Engineers, to investigate greater applications of the Ballistic Launcher in modern warfare and national security.

1.5 Development

The methods followed throughout this investigation will be presented first to provide the reader with a foundation of engineering standards followed. Any assumptions made that may impact the study will be discussed initially. The general methodology followed will be listed before the results of this study to guide the reader through the research investigation. The results will be accompanied with an analysis of the Hardware and Software designs of the Ballistic Launcher apparatus under the 'Discussion' section to justify the success of the design and will additionally allow reconstruction of this research study by the reader. Lastly, conclusions will be drawn from the Ballistic Launcher apparatus created and the Engineering Design Process, and references will be shown to enhance the reader's comprehension on the subject matter.

2.0 Methods, Assumptions, and Procedures

2.1 Methods

All measurements taken were in accordance with the *United States customary units* primarily in inches (in) and feet (ft).

The Ballistic Launcher mechanism was created using standard VEX V5 Parts. In this research study, the ping pong ball was modeled with the use of vex martials along with rubber bands to generate a lift of 15 feet.

The Engineering Process of PLTW was also incorporated to easily generate solutions and follow procedural methodology that could be easily replicated by other parties.

Similar to this the PLTW design matrix was also incorporated to effect create a solution that could be easily built whilst meeting all labeled criteria and constraints of the project.

The VEX V5 Programming platform was used to push programs to the VEX Brain component, acting as a microcontroller for the rest of the machine apparatus. Due to the constraint of time, all construction processes of the Ballistic Launcher apparatus were individually completed through subassemblies to maximize efficiency.

By the same token, all programming processes were completed and tested individually; this enabled the individual functionality of each simple behavior to be tested, facilitating the construction of complex behaviors in the software. Procedural Abstraction was employed to condense complex behaviors into single functions, maximizing the output efficiency of the Ballistic Launcher prototype constructed.

The problem was discovered through a Request for Proposal (RFP), provided by the Directors of Panther Engineering. All research conducted roughly mirrored the methodology in which civil engineering bids are solicited (through RFPs).

2.2 Assumptions



In this research investigation, functionality of the initial prototype was assumed. In the same manner modifications were to be only added and not major changes to the prototype to flow with the continuity and constraints of the initial designer.

It was also assumed that any material in the vex kit and the classroom can be used to modify the Ballistic Launcher.

Similarly, functionality of the VEX V5 Brain was assumed. In order to accurately develop the software for the Ballistic Launcher apparatus, the input from buttons and switches were required. These values were only obtained through the Brain's Digital Display mode. Additionally, the VEX V5 Brain's functionality as a microcontroller was essential to test and correct the software program. Once again, knowledge on the Brain's internal mechanism workings that enable it to perform the aforementioned tasks is beyond the scope of this research investigation; hence, it was assumed to allow the study to continue uninterrupted.

2.3 Procedures

1. Obtain the RFP from the Directors of Panther Engineering and read over the objectives of the project.
2. Create a Design Brief, consolidating the info from the RFP into one document directly for ease of use in the future.
3. Create two initial, working machine designs per person after brainstorming ideas.
4. Create two flowcharts per person, outlining the software design and logic for each machine concept generated in the previous step.
5. Discuss the pros and cons of each design in the team to select favorable traits and ideas. Create a decision matrix, if applicable, to consolidate one final hardware design.
6. Draft a detailed Final Hardware Design (See Appendix D) to model the machine apparatus.
7. Create a detailed Final Software Design (See Appendix E) in the form of a flowchart to model the programming logic and serve as a basis to the final program created.
8. Write a thorough narrative description detailing the mechanical and electrical design of the hardware and the intended software functionality.
9. Gain approval from the Directors of Panther Engineering on the Final Hardware and Software Designs to commence the building process.
10. Create the proposed Hardware Design. Troubleshoot any issues in the construction process by creating workarounds via changing sensor inputs/configuration.
11. Design software to execute the intended functionality of the apparatus. Troubleshoot any issues in the software-design process by using software debugging methods, such as individually running sections of the code or printing variables to the console (in this case, the VEX V5 Brain).
12. Provide a practical demonstration of the device created to the Directors of Panther Engineering to verify that all customer objectives and functionality are satisfied.
13. Draft a Technical Report presenting the findings of the study in accordance with the ANSI/NISO Z39.18-2005 (R2010) Technical Report standards.



3.0 Results and Discussion

3.1 Results

3.1.1 Initial Designs

For team 17, the initial designs of the marble sorter consisted of a variety of ideas utilizing many engineering principles and techniques to efficiently sort the marbles in the time constraint. As a group, team 17 utilized these principles and each group uniquely created a solution based on the marble's unique properties.

The first design (designed by Park; see Appendix B) proposed the use of a gear train for the mobility aspect of the build. Including a potentiometer to adjust the projectile launcher's angle to set angles and an arm restriction embedded into the code to detect if the arm exceeded its physical restriction. Onto the arm of the projectile launcher, the proposed design would use two VEX motors that would use two sprockets connected to the motor to help propel the projectile outwards. With also LED lights on top of the structure as well as a kill switch nearby in a configurable design.

The last design (designed by Park; see Appendix B) proposed the use of gear tracks & rubber bands. The gear tracks would allow a bigger rectangular base to be added onto the base model of the projectile launcher. Though this would add more weight, affecting the design. A rubber band as an effective way to propel an object in a weight effective that would not add any more weight onto the arm of the projectile launcher. With also a wheel to move the adjustable angles of the projectile launcher. The disadvantage of the design is the mobility system is clunky and ineffective, with the rubber band experiencing wear and tear conditions once it is repetitively used.

3.1.2 Proposed Design

The proposed final design (see Appendix D) consisted of a combination of all the ideas presented originally. This design consisted of a gear train and rubber band. The potentiometer adjusts with also the arm restriction embedded, and a kill switch connected to the main arm of the system. The strengths included the different strengths in all the previous initial designs. For example, Park's initial #1 design (see Appendix B) use of the potentiometer, kill switch, and gear train were incorporated in the final design. Park's initial #2 design (see Appendix B) design utilizes a rubber band that helped propelled object afar in greater distances at a low weight aspect of the build as well as the servo motor attached, replacing the VEX motor on the arm of the launcher.

3.1.3 Final As-Build Design

Weight distribution is still evident in the final build, though it does not interfere with the functionality of the projectile launcher. With the completion of the five modifications seamlessly integrated into the base model in an effective manner put together.

3.2 Discussion

Team 17 encountered several problems during the development of the project and provided solutions to them. Team 17 solved these issues by either reinforcing & improving designs or solving them through alternative ways. For instance, Team 17



faced the problem of having modification to the projectile launcher and structural problems to the mechanical build. The adequate solution for the faulty modification of the projectile launcher was to alleviate any wasteful weight added upon the arm of the projectile launcher by using a low-weight solution of rubber bands to launch the projectile. Another problem faced against Team 17 was the attachment of VEX motors onto the projectile launcher system with the weight distribution affecting the whole structure. The team solved the problem by replacing the VEX motor with a Servo motor in the build, adding ease to weight and efficiency to the modular build. Another issue faced by Team 17 was the inconsistent screwing of parts that were regularly falling apart, to solve this issue, each screw was tightened and secured to maintain a sound structural build. In the end, the final build of the ballistic projectile launcher was successful, meeting our RFP objectives.

4.0 Conclusion

Team 17 successfully created a Ballistic Launcher system, meeting the requirements of a 15 feet projectile range, continuous motor motion throughout launch angles from 10 degrees to 80 degrees, and consistent initial velocity through all launch angles. Furthermore, 5 modifications to enhance robot functionality were successfully completed: a joystick was incorporated to ensure robot mobility and adjusting robot aim left and right, a potentiometer was implemented to prevent arm movement beyond physical machine limits and indicate launch angle; and LEDs were incorporated to depict system status. The LED functionalities worked perfectly as the red LED symbolizes that the Ballistic Launcher has gone beyond physical limits, yellow LED symbolizes that the Ballistic Launcher is currently aiming, and the green LED flashes upon firing the projectile. Team 17 learned the importance of organizing and discussing potential solutions by using devices such as the design process and design matrix. Reflections from this project include, most notably, that sensors (e.g. the potentiometer) need to be carefully handled to yield accurate and consistent readings. Moreover, the build needs to be as simple as possible to limit variability and to also increase flexibility for any additional components that need to be added. Lastly, multitasking efforts must be coordinated perfectly, as demonstrated with the novel use of parallel computing devices (two VEX Brains) in Team 17's final Ballistic Launching Device. The final reflection made by Team 17 was that splitting work proved to be useful as all group members applied their strengths to the maximum, allowing the project to be completed comfortably in the time frame given. One lesson learned was that going forward, the ease of mechanical usage will be prioritized as human convenience is just as important to the end users as is the device's success in various use cases.

5.0 References

- Project Lead the Way Engineering.* (n.d.). 3.1.3 Presentation: Program Design Process & Tools Notes. Retrieved April 2, 2024, from <https://classroom.google.com/c/NjE3NjEwNjIzNzQz/a/NjI4NDEzMjM1NTY0/details>
- Project Lead the Way Engineering.* (n.d.). 3.1.4 Presentation: While Loops/If-Else Loops Notes. Retrieved April 2, 2024, from <https://classroom.google.com/c/NjE3NjEwNjIzNzQz/a/NjUxMzgyNzg3OTU2/details>
- Project Lead the Way Engineering.* (n.d.). 3.1.5 Presentation: Variables and Functions. Retrieved April 2, 2024, from <https://classroom.google.com/c/NjE3NjEwNjIzNzQz/a/NjI4NTkwMTYzMDk1/details>



Project Lead the Way Engineering. (n.d.). 4.2.2 Projectile Motion Presentation Notes. Retrieved May 8, 2024, from

<https://classroom.google.com/c/NjE3NjEwNjIzNzQz/a/Njc2NTk5MDg0ODU4/details>

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<https://classroom.google.com/c/NjE3NjEwNjIzNzQz/a/Njc3MDgwMzAyMzcy/details>



Appendices

Appendix A: Design Brief

Table 1: Design Brief

Client	Dauntless Industries, LLC
Target Consumer	Hobby Spectacle, Incorporated
Designer(s)	Aarav Lakdawala, Calvin Park, Vedant Sundar
Problem Statement (restate in your own words)	Hobby Spectacle, Incorporated, has recently discovered that its existing Ballistic Launcher System is inefficient due to energy loss and does not launch the projectile 15 feet far, as desired by customers. This inefficiency is highly prevalent to all industrial consumers requiring the use of the Ballistic Launcher System, specifically military-partnered corporations.
Design Statement	Design, market, and test a modified Ballistic Launcher System based on the existing model by Hobby Spectacle, Incorporated, that provides greater efficiency to stakeholders/customers via an increased ease of use (from a software perspective) and increased range and versatility of projectile deployment (from a hardware perspective).
Criteria: How are the listed criteria measurable?	1. Must be adjustable to different angles (10, 20, 30, 40, 50, 60, 70, and 80 degrees) 2. Must have the same initial velocity, regardless of the launch angle 3. Must launch the projectile (ping-pong ball) at least 15 feet far 4. Must implement 5 design modifications to improve upon design precision, accuracy, and overall structural integrity; additionally, a sensor must be implemented for a closed-loop system 5. Must implement software modifications to complement design modifications and enable greater functionality
Constraints How are the listed constraints measurable?	6. Must be constructed with VEX V5 Parts provided by the Directors of Panther Engineering 7. Design Brief must be submitted for critical review by 4/30 8. Ballistic Design Modification Proposal must be submitted for critical review by 5/1 9. Prototype must be presented via Powerpoint by 5/10 10. Prototype must be demonstrated, along with the submission of a ANSI/NISO Z39 Technical Report, for critical review by 5/10 11. Launch test result data must be recorded for discussion in the final technical report



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Appendix B: Proposed Hardware Design

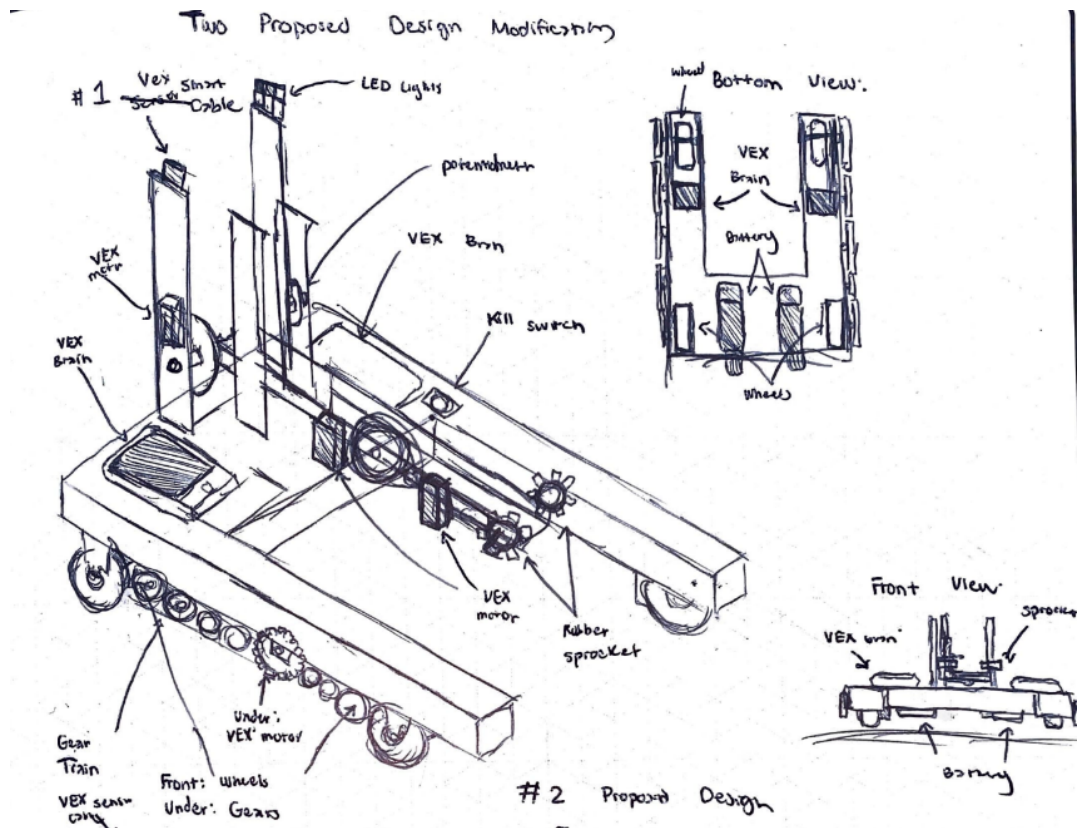


Figure 1: Proposed Hardware Design



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Appendix C: Proposed Software Design

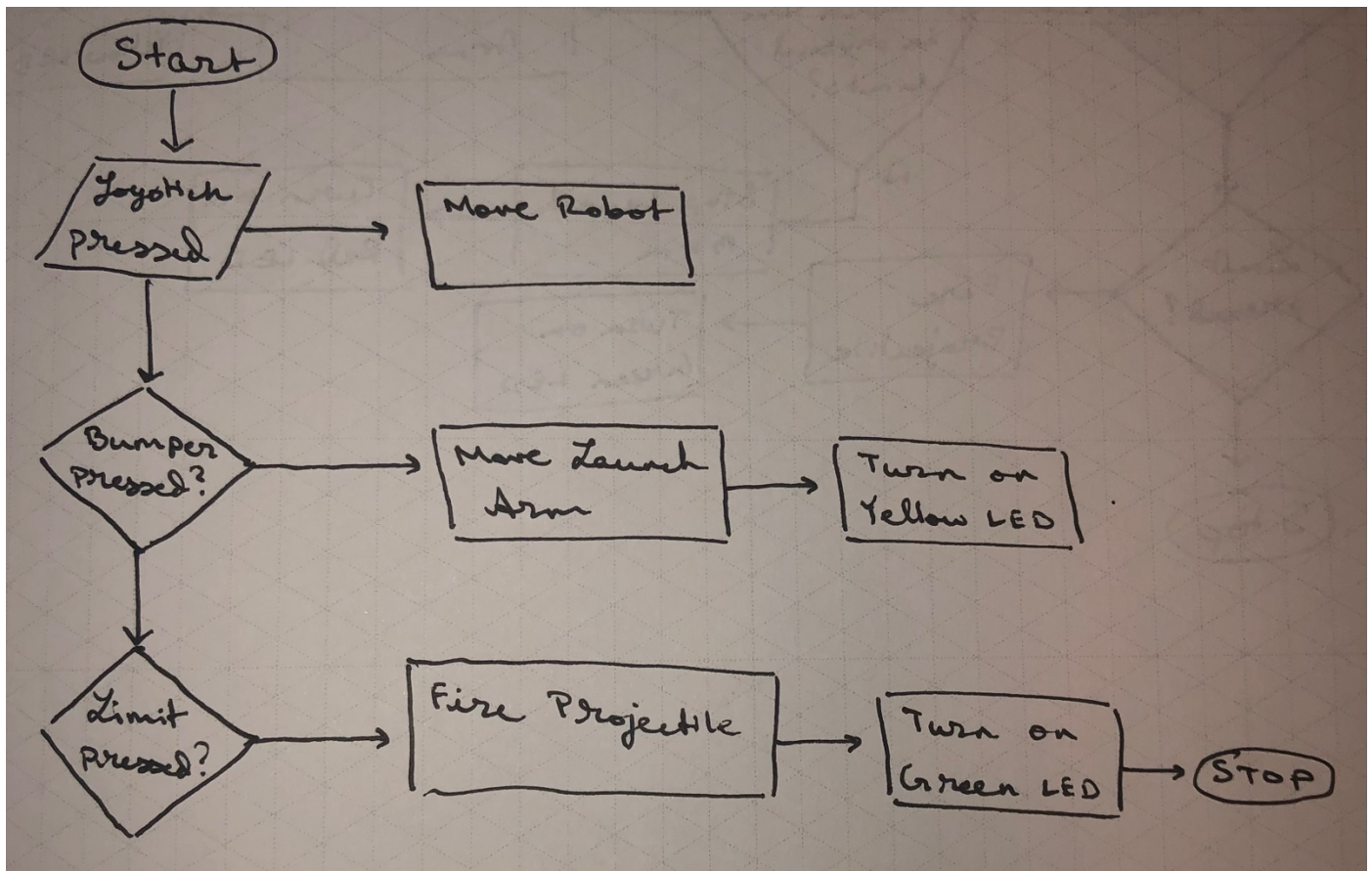


Figure 2: Proposed Software Design



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Appendix D: Final Hardware Design

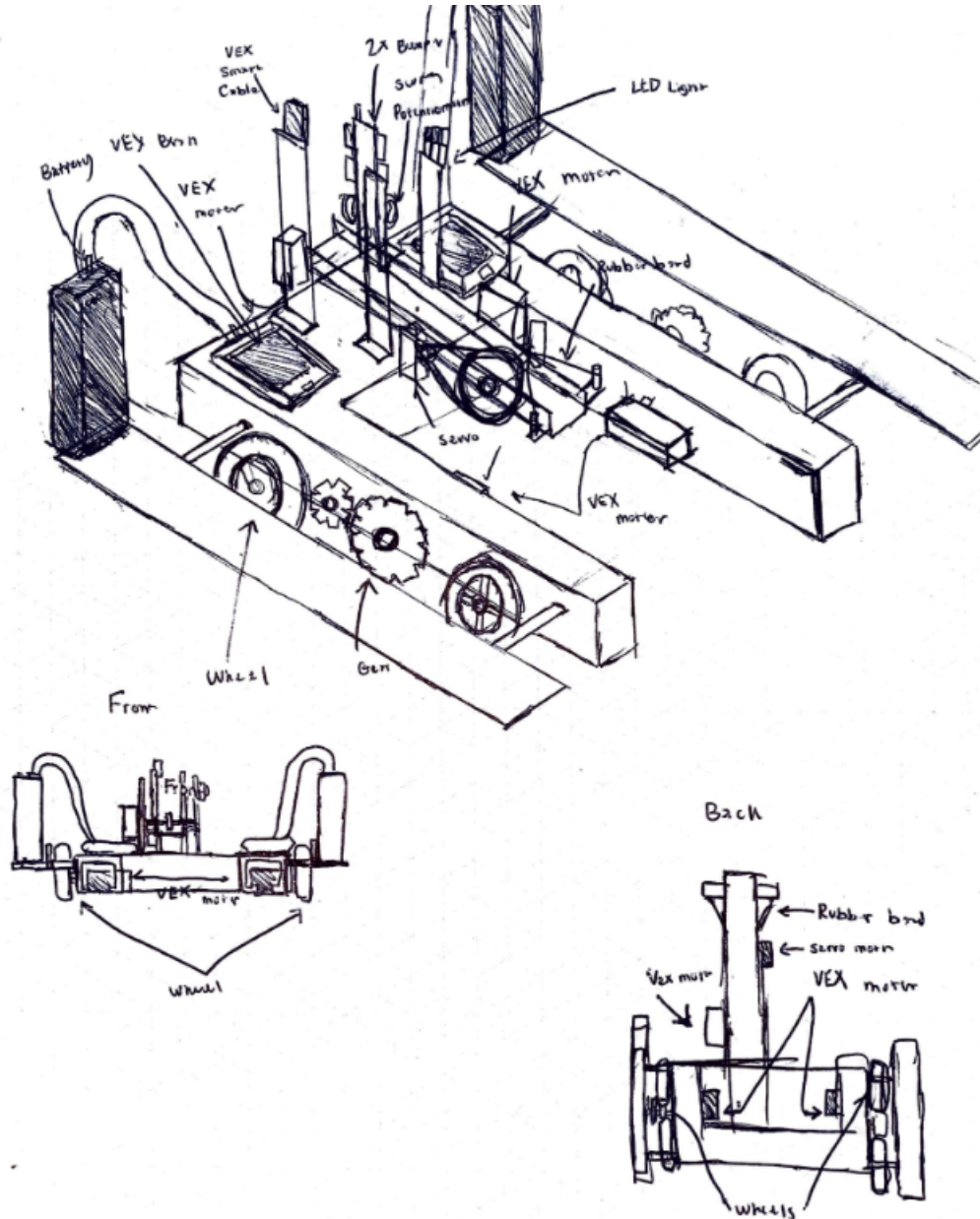


Figure 3: Final Hardware Design



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Appendix E: Final Software Design

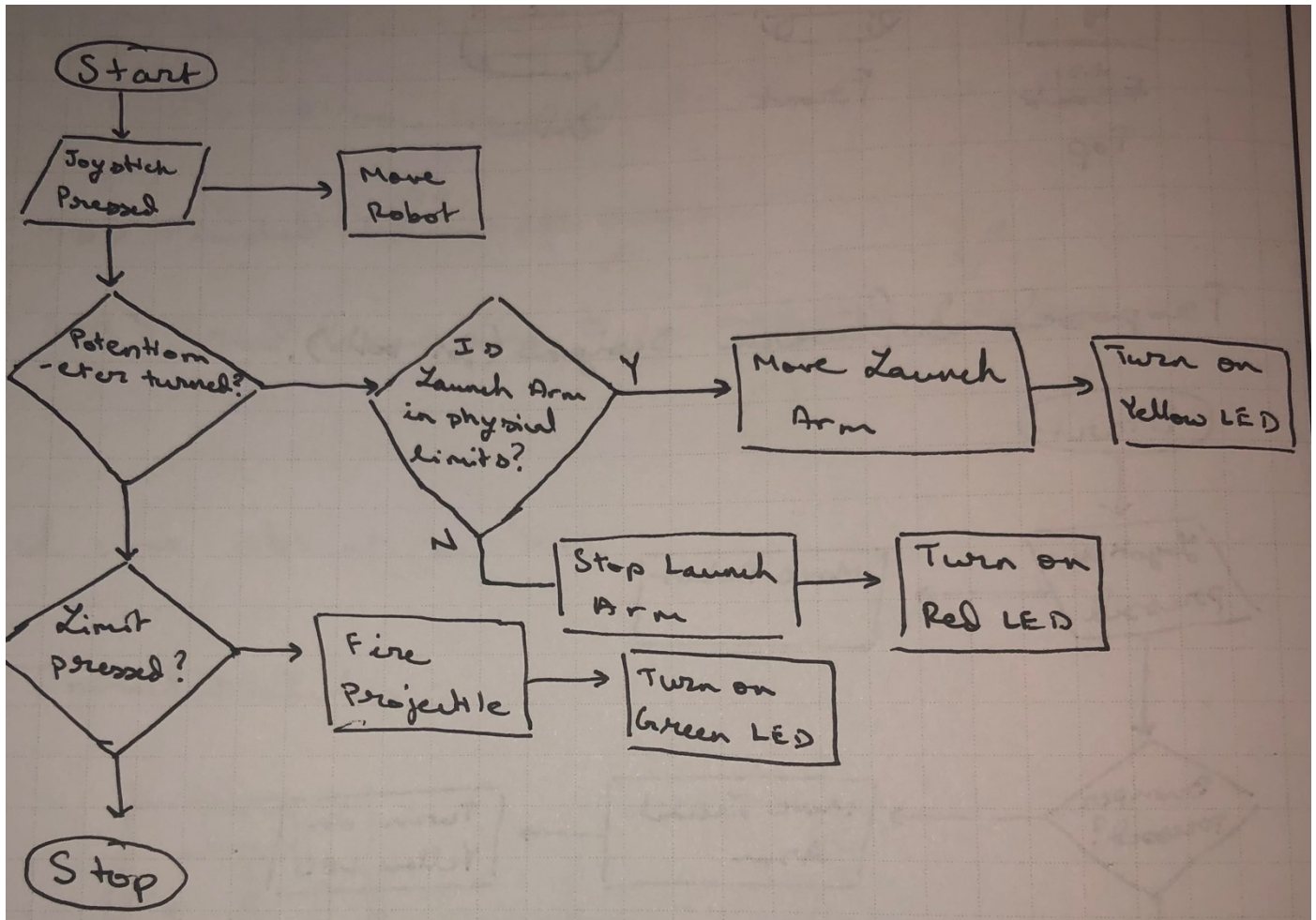


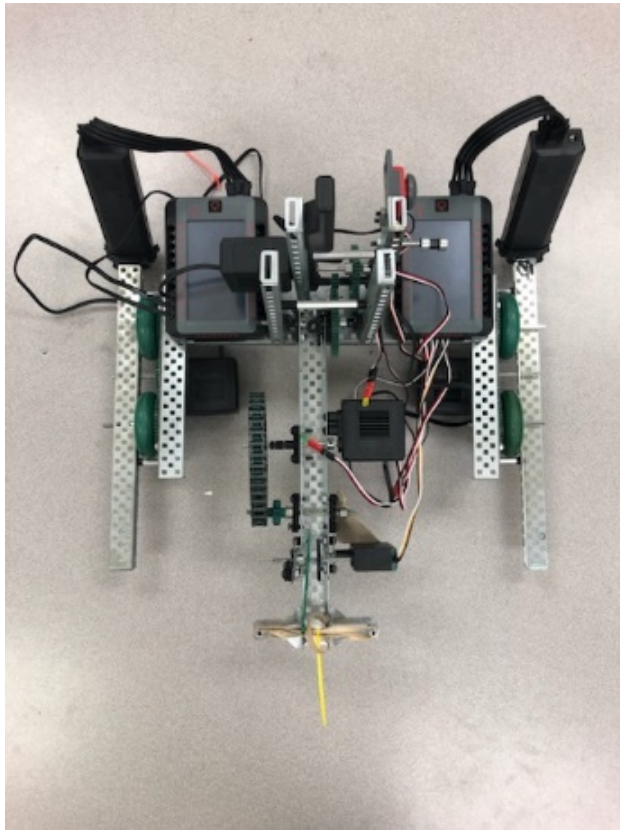
Figure 4: Final Software Design

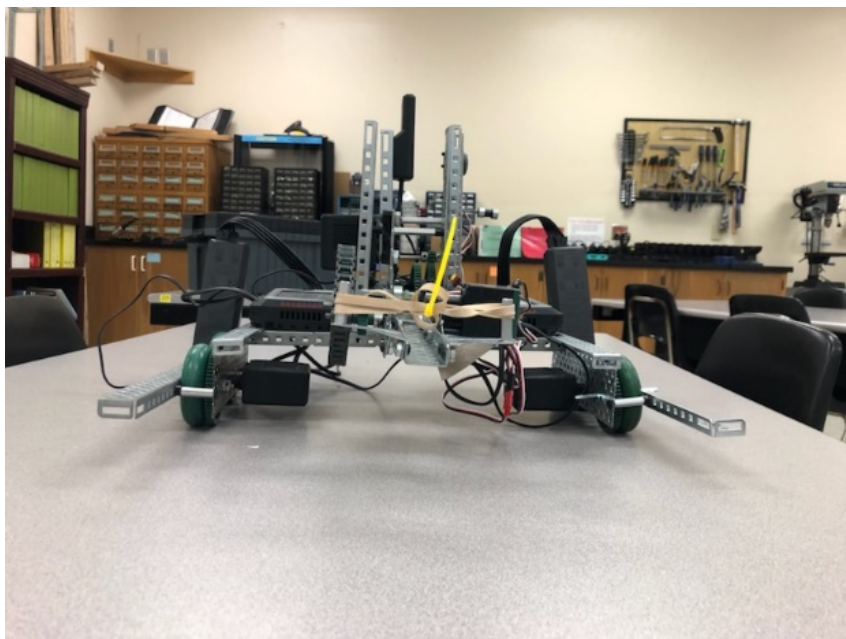


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Appendix F: Photos of Final Build







Appendix G: Final Code

```

vexcode_brain_precision = 0
vexcode_console_precision = 0
ElevSpeed = 0
FireTime = 0
Armed = False

def nullLED():
    global ElevSpeed, FireTime, Armed, vexcode_brain_precision, vexcode_console_precision
    redLEDC.off()
    yellowLEDD.off()
    greenLEDE.off()

def redLED():
    global ElevSpeed, FireTime, Armed, vexcode_brain_precision, vexcode_console_precision
    redLEDC.on()
    yellowLEDD.off()
    greenLEDE.off()

def greenLED():
    global ElevSpeed, FireTime, Armed, vexcode_brain_precision, vexcode_console_precision
    redLEDC.off()
    yellowLEDD.off()
    greenLEDE.on()

def yellowLED():
    global ElevSpeed, FireTime, Armed, vexcode_brain_precision, vexcode_console_precision
    redLEDC.off()
    yellowLEDD.on()
    greenLEDE.off()

def when_started1():
    global ElevSpeed, FireTime, Armed, vexcode_brain_precision, vexcode_console_precision
    servoF.set_position(0 - 50.0, DEGREES)
    Elevation.set_position(0, DEGREES)
    Elevation.set_velocity(50, PERCENT)
    Launch.set_velocity(100, PERCENT)
    ElevSpeed = 100
    nullLED()
    while True:
        if Fire.pressing():
            Launch.set_velocity(100, PERCENT)
            Launch.spin_for(FORWARD, 0.5, TURNS)
            servoF.set_position(100 - 50.0, DEGREES)
            wait(0.1, SECONDS)
            Launch.stop()
            while not not Fire.pressing():
                wait(5, MSEC)

```



```
    nullLED()
while not Fire.pressing():
    if potentiometerB.angle(DEGREES) > 80:
        Elevation.stop()
        brain.screen.print("80")
        brain.screen.next_row()
        redLED()
    else:
        Elevation.spin_to_position((potentiometerB.angle(DEGREES) * -60), DEGREES)
        brain.screen.print(potentiometerB.angle(DEGREES), precision=6 if
vexcode_brain_precision is None else vexcode_brain_precision)
        brain.screen.next_row()
        yellowLED()
        wait(5, MSEC)
        wait(5, MSEC)

when_started1()
```



Appendix H: Projectile Motion Data Sheet

Table 2: Projectile Motion Data Sheet

Table 1: Average Range

Trial Number	Measured Range
1	13.5
2	15
3	18
4	4
5	16
6	8
7	18.5
8	15.5
9	16
10	17
Average Distance	14.15

Calculated v_i

Enter V_i below

26.534

Table 2: Theoretical Range v. Firing Angle

Angle	Range
10°	7.48
20°	14.05
30°	18.94
40°	21.53
45°	21.87
50°	21.53
60°	18.94
70°	14.05
80°	7.48

Table 3: Measured Range v. Firing Angle

Angle	Range 1	Range 2	Range 3	Average Range
10°	11	13	14	12.6666667
20°	15	15.5	10	13.5
30°	16	15	14	15
40°	11	9	14	11.3333333
45°	7	10	10	9
50°	13	11	9	11
60°	7	7	8	7.3333333
70°	6	4	7	5.6666667
80°	4	8	5	5.6666667

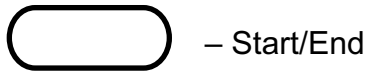


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List of Symbols, Abbreviations, and Acronyms

Symbols



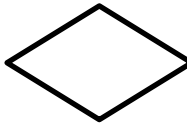
– Start/End



– Process



– Input/Output



– Decision

Abbreviations

VEX

Acronyms

ANSI – American National Standards Institute

ASCE – American Society of Civil Engineers

BIPM – *Bureau International des Poids et Mesures*

LED – Light Emitting Diode

LLC – Limited Liability Company

NISO – National Information Standards Organization

RFP – Request For Proposal

SI – *Système International d'Unités*



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